KPA1500 Thetis Bridge

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The KPA1500 Thetis Bridge is a Windows program that ties an Elecraft KPA1500 amplifier, a FlexRadio FlexControl tuning knob and a Telepost LP-100A wattmeter into Thetis, Thetis HL2 (for the Hermes Lite 2) or SDR-VST3, with meters and buttons inside the program and a browser control panel.

What it does

- **The amp follows the radio.** As you tune Thetis, the bridge sends the frequency to the KPA1500, reads back the frequency the amp reports, and shows whether the two match.
- **Safe tuning.** Its TUNE button puts the amp in STANDBY and confirms it *before* the radio keys, then returns it to OPERATE. Thetis' own TUNE button keys first, so a few milliseconds of RF can reach the amp. The bridge TUNE also stops by itself after 20 seconds.
- **"No antenna?" warning.** If the receive level drops below a limit set per band, the bridge warns before you transmit into nothing. It can also hold the amp in STANDBY until the antenna is back or you press DUMMY OK.
- **Meters and buttons inside Thetis.** Three windows: Telemetry (power, SWR and drive bars with a peak hold, PA temperature, volts, amps, faults), Controls (lit buttons for OPERATE, ATU, antenna, AMP power, TUNE, fan) and Knob. Their layout and titles are set from the bridge, and a red line shows if the bridge stops. They work the same in Thetis, Thetis HL2 and SDR-VST3.
- **FlexControl tuning knob support.** VFO tuning, AF volume, AGC gain, RIT, mute, VOX, band and tuning step. Every press of the knob and its three buttons can be given any of about 30 functions, or a Thetis CAT command of your own. DDUtil and virtual serial ports are not needed.
- **LP-100A wattmeter** readings next to the amp's own.
- **Browser control panel** on the PC, a tablet or a phone. It can also look exactly like the Thetis windows. A password for other devices is optional.

Who it helps and why

It is for users of Thetis, Thetis HL2 and SDR-VST3 (Apache Labs ANAN, Hermes Lite 2 and other openHPSDR radios) who want their station hardware to work together without a stack of separate utilities.

- **KPA1500 owners** get band tracking, Standby-on-Tune without the RF spike at key-up, and amp meters and controls on the Thetis screen. They no longer need the Elecraft utility open beside Thetis.
- **Anyone with a FlexControl knob** gets a real tuning knob in Thetis, with or without the amplifier, that can be set up to their own taste.
- **The station is protected:**
  - TUNE never keys into an amp that is still in OPERATE.
  - Antenna and ATU changes are refused under RF.
  - Power-off under RF can be blocked.
  - The no-antenna check catches a wrong antenna switch or a loose cable before transmit.
- **Remote and shack-wide use:** amp status and controls from any browser on the home network.
- **Non-programmers are covered:** a one-click installer, built-in help with pictures, and settings changed from the panel rather than in files.

What you need and how to start

Item	Needed
Computer	Windows 10 or 11
Thetis	Thetis 2.10.3.12 or newer, Thetis HL2 2.10.3.15 or newer, or SDR-VST3 5.x, with its TCI server switched on (Setup > Serial/Network/Midi CAT > Network, 127.0.0.1:50001)
Python	3.8 or newer (the installer offers to install it)
Elecraft KPA1500	Optional, on the home network
FlexRadio FlexControl knob	Optional, USB

Item	Needed
Telepost LP-100A	Optional, USB

1. Unzip **KPA1500_Bridge_Complete_v231.zip** and double-click **1_RUN_THIS_Installer.bat**.
2. Answer the questions; pressing Enter accepts the suggested answer. The bridge goes into its own folder, C:\RadioScripts\KPA1500_Thetis_Bridge.
3. Let it add the windows to Thetis, Thetis HL2 or SDR-VST3: one, several or all. It closes a running program the normal way first and backs up each program's settings; Undo_Thetis_Containers.bat puts them back.
4. Start the bridge from the Desktop shortcut, then the program (the start file can launch it for you). The panel is at http://127.0.0.1:8080, and the full help opens from it.

The programs' own files are never changed. Only their settings files get the new windows, after a backup. Only one of the three runs at a time, since they share TCI port 50001 and the radio. Version 2.31. Concept, design and on-air testing: KW4EX.